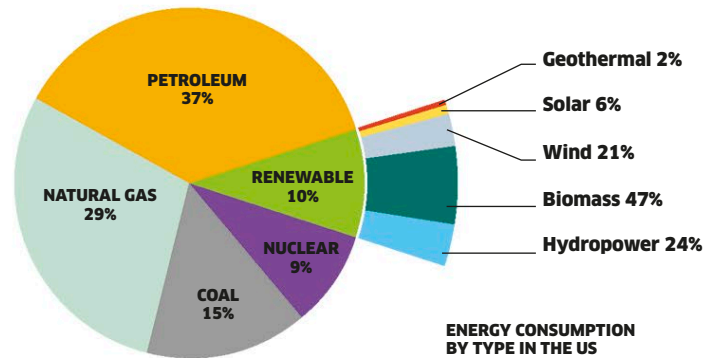


ENERGY SOURCES

People in the industrialized world use a lot of energy in homes, business, and industry, for travel and transportation. The energy used comes from primary sources such as fossil fuels, nuclear energy, and hydropower. Crude oil, natural gas, and coal are called fossil fuels because they were formed over millions of years by heat from Earth's core and pressure from rock on the remains (fossils) of plants and animals (see p.37).

Energy consumption

Most energy consumed in the US is from nonrenewable sources, with more than 80 percent derived from fossil fuels. Despite advances, just 10 percent comes from renewable sources, of which nearly half is from biomass.



Nonrenewable sources

Fossil fuels are limited resources on our planet, which create greenhouse gases (see pp.128–129) and toxic pollutants. Nuclear energy produces fewer greenhouse gases, but leaves harmful waste.



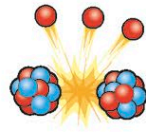
Crude oil
Liquid hydrocarbons found deep underground.



Natural gas
Hydrocarbon gas formed millions of years ago.



Coal
Solid hydrocarbons made by heat and pressure.



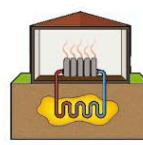
Nuclear energy
Energy released by splitting uranium atoms.

Renewable sources

Energy produced by resources that cannot run out, such as sunlight, wind, and water, is more sustainable. Their use does not produce greenhouse gases and other harmful waste products. Biomass releases carbon dioxide, however, and must be offset by planting new trees.



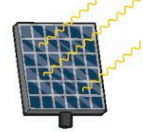
Biomass
Fuel from wood, plant matter, and waste.



Geothermal energy
Heat deep inside the Earth, in water and rock.



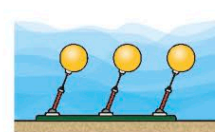
Wind power
Moving air caused by uneven heating of Earth.



Solar energy
The sun's radiation, converted into heat.



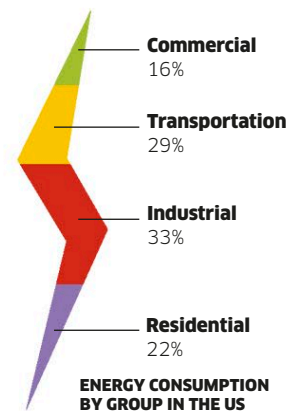
Hydropower
The energy of falling or flowing water.



Tidal and wave power
The motion of tides and wind-driven waves.

Energy use

In the developed world, industry and transportation are the most energy-hungry sectors, while efficiency has reduced energy consumption in the home.



ELECTRICAL GRID

Regardless of the primary energy source, most energy is delivered to users as electrical energy. The network of cables used to distribute electricity to homes, offices, and factories is called the electrical grid. Many sources, including wind and solar, feed into the grid, but the majority of electricity is generated in power stations, which use the energy released by burning fossil fuels to power huge electrical generators.

